

4.11.6 Data Visualization in MS Access (Charts)

بار لمبے ناموں والی اقسام، کالم سادہ موازنہ، پائی حصوں کا تناسب، لائن وقت کے ساتھ رجحان۔

CS XI

Chapter 4: Data and Analysis

Lecture 15 of 15

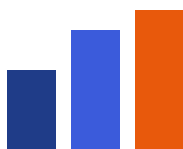
4.11.6 Data Visualization in MS Access (Charts)

LEARN IT

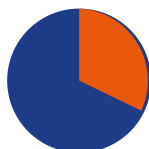
Charts turn numbers into a picture so a pattern jumps out. Pick the chart that matches the question:

Chart	Shape	Best for
Bar (horizontal)	Bars sideways	Categories with long names (English, Mathematics, ...)
Column (vertical)	Bars standing	Side-by-side comparison (2024 / 2025 / 2026 admissions)
Pie	Slices of a circle	Parts of one whole (percent of CS vs IT vs Eng). Few slices.
Line	Dots joined	A trend over time (visitors per month)

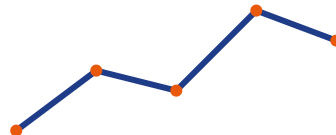
Column



Pie



Line (trend)



Column = compare categories. Pie = share of a total. Line = change over time.

EXAM TIP

"Which chart?" — line for time, pie for composition, bar/column for comparing categories. Mention one reason.

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CHECK YOURSELF

1. Website visitors each month — which chart?
2. Share of boys vs girls in one class — which chart?

Answers: 1. Line. 2. Pie (two parts of a whole).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A column chart in Access is for:

- a. Comparing categories
- b. Parts of a whole only
- c. Time only
- d. OSI only

MCQ 2. A pie chart shows:

- a. Share of a total
- b. A trend of 60 months well
- c. Binary search
- d. Foreign keys

MCQ 3. A line chart shows:

- a. Change over time
- b. Unrelated categories best
- c. ER diamonds
- d. NAND

MCQ 4. Charts should be based on:

- a. A summarised query, not 10,000 raw rows if possible
- b. The relationship window
- c. Python turtle
- d. Figma

MCQ 5. Unlabelled axes in an exam chart score:

- a. Near zero
- b. Full marks
- c. Bonus NAND
- d. An extra sprint

MCQ 6. Choose the chart that matches the question, same rule as:

- a. Matplotlib in XII
- b. Bubble sort
- c. TCP handshake
- d. K-maps

Extra short questions

1. When pie vs column?
2. When line?
3. Why chart a totals query not the raw table?
4. Three labels every chart needs.

Extra long question

1. For library data pick a chart: (a) books per genre (b) issues per month (c) share of lost vs on-shelf vs borrowed. Justify and sketch.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Pie: composition of one whole (few slices). Column: compare separate categories.
2. Enrolment 2019–2026 — time.
3. The chart needs one number per category, not every student.
4. Title, axis titles / legend, units.

Long

1. (a) column (b) line (c) pie. Each sketch labelled. One sentence why the others are worse.

Next lecture: Chapter exam (in the chapter lecture PDF)